Integration by Parts

Integration by parts is a technique for performing indefinite integration $\int u\,dv$ or definite integration $\int_a^b u\,dv$ by expanding the differential of a product of functions $d(uv)$ and expressing the original integral in terms of a known integral $\int v\,du$. A single integration by parts starts with

$$d(uv) = u\,dv + v\,du, \tag{1}$$

and integrates both sides,

$$\int d(uv) = uv = \int u\,dv + \int v\,du. \tag{2}$$

Rearranging gives

$$\int u\,dv = uv - \int v\,du. \tag{3}$$

For example, consider the integral $\int x\cos x\,dx$ and let

$$u = x \quad dv = \cos x\,dx \tag{4}$$

$$du = dx \quad v = \sin x, \tag{5}$$

so integration by parts gives

$$\int x\cos x\,dx = x\sin x - \int \sin x\,dx \tag{6}$$

$$= x\sin x + \cos x + C, \tag{7}$$

where C is a constant of integration.

The procedure does not always succeed, since some choices of u may lead to more complicated integrals than the original. For example, consider again the integral $\int x\cos x\,dx$ and let

$$\begin{aligned} u &= \cos x \quad dv = x\,dx \\ du &= -\sin x\,dx \quad v = \tfrac{1}{2}x^2, \end{aligned} \tag{8}$$

giving

$$\int x\cos x\,dx = \tfrac{1}{2}x^2\cos x - \tfrac{1}{2}\int x^2(-\sin x)\,dx \tag{9}$$

$$= \tfrac{1}{2}x^2\cos x + \tfrac{1}{2}\int x^2\sin x\,dx, \tag{10}$$

which is more difficult than the original (Apostol 1967, pp. 218-219).

Integration by parts may also fail because it leads back to the original integral. For example, consider $\int x^{-1}\,dx$ and let

$$\begin{aligned} u &= x \quad dv = x^{-2}\,dx \\ du &= dx \quad v = -x^{-1}, \end{aligned} \tag{11}$$

then

$$\int x^{-1}\,dx = -1 - \int\left(-x^{-1}\right)dx + C = \int x^{-1}\,dx + C - 1, \tag{12}$$

which is same integral as the original (Apostol 1967, p. 219).

The analogous procedure works for definite integration by parts, so

$$\int_a^b u\,dv = [uv]_a^b - \int_a^b v\,du, \tag{13}$$

where $[f]_a^b = f(b) - f(a)$.

Integration by parts can also be applied n times to $\int f^{(n)}(x)\,g(x)\,dx$:

$$\begin{aligned} u &= g(x) \quad dv = f^{(n)}(x)\,dx \\ du &= g'(x)\,dx \quad v = f^{(n-1)}(x). \end{aligned} \tag{14}$$

Therefore,

$$\int f^{(n)}(x)\,g(x)\,dx = g(x)\,f^{(n-1)}(x) - \int f^{(n-1)}(x)\,g'(x)\,dx. \tag{15}$$

But

$$\int f^{(n-1)}(x)\,g'(x)\,dx = g'(x)\,f^{(n-2)}(x) - \int f^{(n-2)}(x)\,g''(x)\,dx \tag{16}$$

$$\int f^{(n-2)}(x)\,g''(x)\,dx = g''(x)\,f^{(n-3)}(x) - \int f^{(n-3)}(x)\,g^{(3)}(x)\,dx, \tag{17}$$

so

$$\int f^{(n)}(x)\,g(x)\,dx = g(x)\,f^{(n-1)}(x) - g'(x)\,f^{(n-2)}(x) + g''(x)\,f^{(n-3)}(x) - \ldots + (-1)^n \int f(x)\,g^{(n)}(x)\,dx. \tag{18}$$

Now consider this in the slightly different form $\int f(x)\,g(x)\,dx$. Integrate by parts a first time

$$\begin{aligned} u &= f(x) \quad dv = g(x)\,dx \\ du &= f'(x)\,dx \quad v = \int g(x)\,dx, \end{aligned} \tag{19}$$

so

$$\int f(x)\,g(x)\,dx = f(x)\int g(x)\,dx - \int\left[\int g(x)\,dx\right] f'(x)\,dx. \tag{20}$$

Now integrate by parts a second time,

$$\begin{aligned} u &= f'(x) \quad dv = \int g(x)\,dx \\ du &= f''(x)\,dx \quad v = \iint g(x)\,(dx)^2, \end{aligned} \tag{21}$$

so

$$\int f(x)\,g(x)\,dx = f(x)\int g(x)\,dx - f'(x)\iint g(x)\,(dx)^2 + \int\left[\iint g(x)\,(dx)^2\right] f''(x)\,dx. \tag{22}$$

Repeating a third time,

$$\int f(x)\,g(x)\,dx = f(x)\int g(x)\,dx - f'(x)\iint g(x)\,(dx)^2 + f''(x)\iiint g(x)\,(dx)^3 - \int\left[\iiint g(x)\,(dx)^3\right] f'''(x)\,dx. \tag{23}$$

Therefore, after n applications,

$$\int f(x)\,g(x)\,dx = f(x)\int g(x)\,dx - f'(x)\iint g(x)\,(dx)^2 + f''(x)\iiint g(x)\,(dx)^3 - \ldots + (-1)^{n+1} f^{(n)}(x)\underbrace{\int\cdots\int}_{n+1} g(x)\,(dx)^{n+1} + (-1)^n \int\left[\underbrace{\int\cdots\int}_{n+1} g(x)\,(dx)^{n+1}\right] f^{(n+1)}(x)\,dx. \tag{24}$$

If $f^{(n+1)}(x) = 0$ (e.g., for an nth degree polynomial), the last term is 0, so the sum terminates after n terms and

$$\int f(x)\,g(x)\,dx = f(x)\int g(x)\,dx - f'(x)\iint g(x)\,(dx)^2 + f''(x)\iiint g(x)\,(dx)^3 - \ldots + (-1)^{n+1} f^{(n)}(x)\underbrace{\int\cdots\int}_{n+1} g(x)\,(dx)^{n+1}. \tag{25}$$

SEE ALSO

Darboux's Formula, Integral, Integration, Inverse Function Integration, Summation by Parts

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integration by parts =

More things to try: = integration by parts | = 8_1 knot | = d/dz am(z, m)

REFERENCES

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